A practitioner's guide to the **cS** package

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This guide shows how to implement the contaminated survival (cS) model with the **cS** package for R. It reproduces the results for the Ecuadorian birth registration data in Otto (2026) and shows how to apply the model to your own data. The package is available on GitHub at https://github.com/arnootto/cS.

The cS model is a two-component mixture for time-to-event data in which both components share the same reference distribution and differ only in speed. A proportion α of the observations belongs to a *delayed* component whose hazard is multiplied by $\eta \in (0, 1)$, so these observations take longer to experience the event. The model was developed to identify delayed birth registrations from two administrative dates: the date of birth and the date of registration. It tells you

1. **how many** registrations follow a slower regime (α),
2. **how much slower** that regime is (η), and
3. **which** records are likely to belong to it (a posterior probability for every birth).

Concomitant variables can be added to see **which groups** are more likely to be delayed. Two reference distributions are implemented: the exponential (giving the contaminated exponential, cE) and the Weibull (giving the contaminated Weibull, cW).

Installation

To install the **cS** package directly from GitHub, use the following code in R:

```
# install.packages("devtools")
library(devtools)
install_github("arnootto/cS")
library(cS)
```

The model

The survival function of the cS model is

$$S_{\mathsf{C}}(t) = (1 - \alpha)\, S_{\mathsf{R}}(t) + \alpha\, S_{\mathsf{R}}(t)^{\eta},$$

where $S_{\mathsf{R}}(t)$ is the reference survival function, $\alpha \in (0, 1)$ is the proportion of observations in the delayed component, and $\eta \in (0, 1)$ is the hazard multiplier of that component. For the contaminated Weibull, every quantile of the delayed component (the median, for example) is $\eta^{-1/k}$ times the corresponding quantile of the reference component. The functions return this as `time.ratio`.

With concomitant variables $\boldsymbol{x}_i$, the proportion becomes

$$\alpha_i = \frac{\exp(\boldsymbol{x}_i'\boldsymbol{\gamma})}{1 + \exp(\boldsymbol{x}_i'\boldsymbol{\gamma})}.$$

All models are fitted by maximum likelihood with the EM algorithm.

Main functions

`fit.e()`, `fit.w()`	Fit the exponential and Weibull reference models.
`ml.cexp()`, `ml.cweib()`	Fit the cE and cW models.
`ml.cexp.cv()`, `ml.cweib.cv()`	Fit the cE and cW models with concomitant variables.
`lrt()`	Likelihood ratio test between two nested concomitant models.
`births`	January 2024 Ecuadorian birth registration data.

Example: Ecuadorian birth registrations

The `births` data contain the 17,144 live births that occurred in Ecuador in January 2024 (INEC, 2024), with the date of birth, the date of registration, the lag in days, and four indicators (`home`, `rural`, `indigenous`, `out_province`). The code below reproduces the results in Otto (2026).

Step 1: Prepare the survival time

The registry records whole days, and some births are registered on the day of birth (`lag = 0`). We use the midpoint of the day, `lag + 0.5`.

```
library(cS)
data(births)
births$t <- births$lag + 0.5
```

Step 2: Is there a delayed regime?

Fit the two reference models and their contaminated versions, and compare them with BIC (lower is better).

```
fe  <- fit.e(births$t)
fw  <- fit.w(births$t)
fce <- ml.cexp(births$t)
fcw <- ml.cweib(births$t)

c(E = fe$BIC, W = fw$BIC, cE = fce$BIC, cW = fcw$BIC)
#>        E        W       cE       cW
#> 152687.7 150858.6 149143.3 148943.4
```

Both contaminated models fit far better than their reference models, and the cW fits best.

Step 3: How large and how slow is the delayed regime?

```
fcw$results
#>       estimates    se
#> sigma    14.661 0.227
#> k         1.134 0.009
#> alpha     0.270 0.008
#> eta       0.140 0.004

fcw$time.ratio
#> 5.67
```

About 27% of births belong to the delayed regime, and every quantile of the delayed component is about 5.7 times the corresponding quantile of the routine component. The routine median is $\hat{\sigma}(\ln 2)^{1/\hat{k}} \approx 10.6$ days, and the delayed median is about $5.67 \times 10.6 \approx 60$ days.

Step 4: Which births are delayed?

`fcw$v` holds the posterior probability that each birth belongs to the delayed component. Classify a birth as delayed if this probability exceeds 0.5.

```
births$prob.delayed <- fcw$v
births$delayed <- ifelse(fcw$v > 0.5, 1, 0)
table(births$delayed)
#>     0     1
#> 13811  3333
```

Step 5: Which groups are more likely to be delayed?

Let the proportion α depend on the four indicators, and test whether they improve the fit.

```
f0 <- ml.cweib.cv(t ~ 1, ~ 1, data = births)
f1 <- ml.cweib.cv(t ~ 1, ~ home + rural + indigenous + out_province,
                  data = births)

lrt(f0, f1)
#>    loglike0  loglike1     stat df             p
#> 1 -74452.22 -74198.71 507.0136  4 2.037444e-108

f1$results
exp(f1$gamma[-1])     # odds ratios for belonging to the delayed component
#>         home        rural   indigenous out_province
#>         6.27         1.35         2.13         8.15
```

A birth registered outside its province of birth has about eight times the odds of belonging to the delayed component, and a home birth about six times. `f1$alpha` gives the fitted probability of delayed registration for every birth.

Using your own data

All the model needs is two dates per birth. If your data frame `mydata` has columns `birth_date` and `registration_date`:

```
mydata$birth_date        <- as.Date(mydata$birth_date)
mydata$registration_date <- as.Date(mydata$registration_date)

mydata$lag <- as.numeric(mydata$registration_date - mydata$birth_date)
mydata <- mydata[mydata$lag >= 0, ]
mydata$t <- mydata$lag + 0.5

fit <- ml.cweib(mydata$t)
fit$results
fit$time.ratio
```

Then follow Steps 2–5 above, replacing `births` with `mydata`.

Return values

`ml.cexp()` and `ml.cweib()` return a list with, among others:

`results`	Estimates and standard errors.
`alpha`, `eta`	Proportion delayed and hazard multiplier.
`lambda` (cE) or `sigma`, `k` (cW)	Parameters of the reference distribution.

`time.ratio`	How many times slower the delayed component is.
`v`	Posterior probability that each observation is delayed.
`loglike`, `AIC`, `BIC`	Log-likelihood and information criteria.
`AIC.ref`, `BIC.ref`	Information criteria of the reference model.

`ml.cexp.cv()` and `ml.cweib.cv()` also return `gamma` (the coefficients for α, on the logit scale), `alpha` (the fitted proportion for every observation) and `npar` (the number of parameters, used by `lrt()`).

Other functions in the cS package

`dcexp()`, `scexp()`, `hcexp()`	Density, survival and hazard of the cE distribution.
`dcweib()`, `scweib()`, `hcweib()`	Density, survival and hazard of the cW distribution.
`h0weib()`	Hazard of the reference Weibull distribution.
`pi.cexp()`	Proportion of delayed observations still at risk at time t (cE).
`rcexp()`, `rcweib()`	Simulate from the cE and cW distributions.
`rcexp.cv()`, `rcweib.cv()`	Simulate from the models with concomitant variables.

For more details, use the `help` command or type `?function_name` in `R` (e.g., `?ml.cweib`).

References

Instituto Nacional de Estadística y Censos (INEC) (2024). Estadísticas de Nacidos Vivos 2024 (ENV). Open data, https://www.ecuadorencifras.gob.ec.

Otto, A. (2026). Who registers late? Identifying delayed birth registrations from two administrative dates.